Project Title	**Netflix Data: Cleaning, Analysis and Visualization**
Tools	Python, ML, SQL, Excel
Domain	Data Analyst & Data scientist
Project Difficulties level	intermediate

Dataset : Dataset is available in the given link. You can download it at your convenience.

Click here to download data set

About Dataset

Netflix is a popular streaming service that offers a vast catalog of movies, TV shows, and original contents. This dataset is a cleaned version of the original version which can be found here. The data consist of contents added to Netflix from 2008 to 2021. The oldest content is as old as 1925 and the newest as 2021. This dataset will be cleaned with PostgreSQL and visualized with Tableau. The purpose of this dataset is to test my data cleaning and visualization skills. The cleaned data can be found below and the Tableau dashboard can be found here .

Data Cleaning

We are going to

1. Treat the Nulls
2. Treat the duplicates
3. Populate missing rows
4. Drop unneeded columns
5. Split columns

   Extra steps and more explanation on the process will be explained through the code comments

Example: You can get the basic idea how you can create a project from here

Netflix Data: Cleaning, Analysis, and Visualization (Beginner ML Project)

This project involves loading, cleaning, analyzing, and visualizing data from a Netflix dataset. We'll use Python libraries like Pandas, Matplotlib, and Seaborn to work through the project. The goal is to explore the dataset, derive insights, and prepare for potential machine learning tasks.

Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
```

Step 2: Load the Dataset

Assume we have a dataset named `netflix_titles.csv`.

```
# Load the dataset
data = pd.read_csv('netflix_titles.csv')

# Display the first few rows of the dataset
print(data.head())
```

Step 3: Data Cleaning

Identify and handle missing data, correct data types, and drop duplicates.

```
# Check for missing values
print(data.isnull().sum())

# Drop duplicates if any
data.drop_duplicates(inplace=True)

# Drop rows with missing critical information
data.dropna(subset=['director',        'cast',        'country'],
inplace=True)

# Convert 'date_added' to datetime
data['date_added'] = pd.to_datetime(data['date_added'])
```

```
# Show data types to confirm changes
print(data.dtypes)
```

Step 4: Exploratory Data Analysis (EDA)

1. Content Type Distribution (Movies vs. TV Shows)

```
# Count the number of Movies and TV Shows
type_counts = data['type'].value_counts()

# Plot the distribution
plt.figure(figsize=(8, 6))
sns.barplot(x=type_counts.index,          y=type_counts.values,
palette='Set2')
plt.title('Distribution of Content by Type')
plt.xlabel('Type')
plt.ylabel('Count')
plt.show()
```

2. Most Common Genres

```
# Split the 'listed_in' column and count genres
data['genres'] = data['listed_in'].apply(lambda x: x.split(',
'))
all_genres = sum(data['genres'], [])
genre_counts = pd.Series(all_genres).value_counts().head(10)
```

```
# Plot the most common genres
plt.figure(figsize=(10, 6))
sns.barplot(x=genre_counts.values,        y=genre_counts.index,
palette='Set3')
plt.title('Most Common Genres on Netflix')
plt.xlabel('Count')
plt.ylabel('Genre')
plt.show()
```

3. Content Added Over Time

```
# Extract year and month from 'date_added'
data['year_added'] = data['date_added'].dt.year
data['month_added'] = data['date_added'].dt.month

# Plot content added over the years
plt.figure(figsize=(12, 6))
sns.countplot(x='year_added', data=data, palette='coolwarm')
plt.title('Content Added Over Time')
plt.xlabel('Year')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.show()
```

4. Top 10 Directors with the Most Titles

```
# Count titles by director
top_directors = data['director'].value_counts().head(10)

# Plot top directors
plt.figure(figsize=(10, 6))
sns.barplot(x=top_directors.values,      y=top_directors.index,
palette='Blues_d')
plt.title('Top 10 Directors with the Most Titles')
plt.xlabel('Number of Titles')
plt.ylabel('Director')
plt.show()
```

5. Word Cloud of Movie Titles

```
# Generate word cloud
movie_titles = data[data['type'] == 'Movie']['title']
wordcloud       =       WordCloud(width=800,        height=400,
background_color='black').generate(' '.join(movie_titles))

# Plot word cloud
plt.figure(figsize=(10, 6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.show()
```

Step 5: Conclusion and Insights

In this project, we:

1. **Cleaned the data** by handling missing values, removing duplicates, and converting data types.
2. **Explored the data** through various visualizations such as bar plots and word clouds.
3. **Analyzed content trends** over time, identified popular genres, and highlighted top directors.

Step 6: Next Steps

1. **Feature Engineering**: Create new features, such as counting the number of genres per movie or extracting the duration in minutes.
2. **Machine Learning**: Use the cleaned and processed data to build models for recommendations or trend predictions.
3. **Advanced Visualization**: Use interactive plots or dashboards for more detailed analysis.

This project is a foundational exercise that introduces essential data analysis techniques, paving the way for more advanced projects.

Sample code

```
import pandas as pd
import numpy as np
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

Importing data from csv and getting info about data.

In [2]:

```
data=pd.read_csv("/kaggle/input/netflix-data-cleaning-analysis-and-visualization/netflix1.csv")
data.head()
```

Out[2]:

	show_id	type	title	director	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	United States	9/25/2021	2020	PG-13	90 min	Documentaries
1	s3	TV Show	Ganglands	Julien Leclercq	France	9/24/2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...

2	s6	TV Show	Midnight Mass	Mike Flanagan	United States	9/24/2021	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries
3	s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Brazil	9/22/2021	2021	TV-PG	91 min	Children & Family Movies, Comedies
4	s8	Movie	Sankofa	Haile Gerima	United States	9/24/2021	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies

In [3]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8790 non-null   object
 1   type          8790 non-null   object
 2   title         8790 non-null   object
 3   director      8790 non-null   object
 4   country       8790 non-null   object
```

```
 5   date_added    8790 non-null   object
 6   release_year  8790 non-null   int64
 7   rating        8790 non-null   object
 8   duration      8790 non-null   object
 9   listed_in     8790 non-null   object
dtypes: int64(1), object(9)
memory usage: 686.8+ KB
```

In [4]:

```
data.shape
```

Out[4]:

```
(8790, 10)
```

In [5]:

```
data=data.drop_duplicates()
```

Content distribution on Netflix.

In [6]:

```
data['type'].value_counts()
```

Out[6]:

```
type
Movie       6126
```

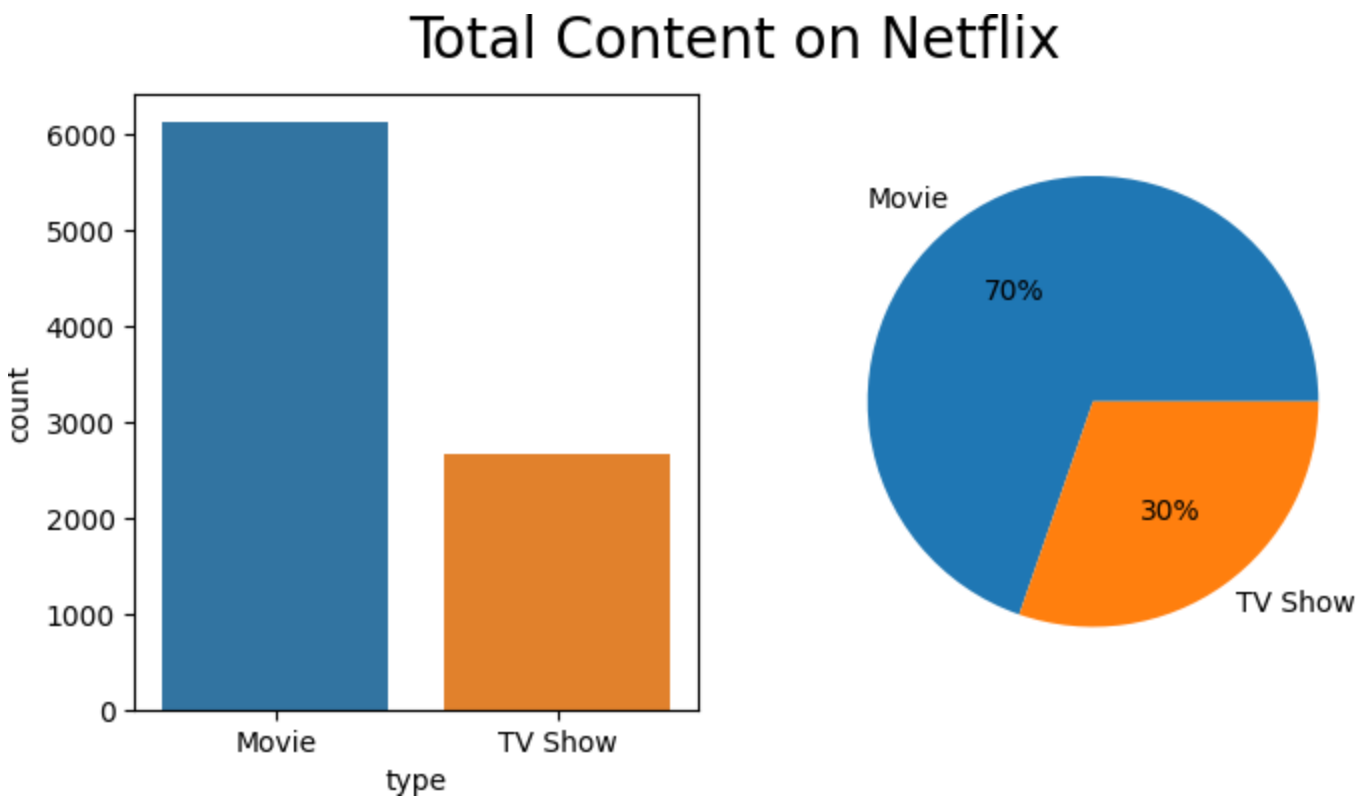
```
TV Show    2664

Name: count, dtype: int64
```

In [7]:

```
freq=data['type'].value_counts()

fig, axes=plt.subplots(1,2, figsize=(8, 4))

sns.countplot(data, x=data['type'], ax=axes[0])
plt.pie(freq, labels=['Movie', 'TV Show'], autopct='%.0f%%')

plt.suptitle('Total Content on Netflix', fontsize=20)
```

Out[7]:

```
Text(0.5, 0.98, 'Total Content on Netflix')
```



In [8]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8790 non-null   object
 1   type          8790 non-null   object
 2   title         8790 non-null   object
 3   director      8790 non-null   object
 4   country       8790 non-null   object
 5   date_added    8790 non-null   object
 6   release_year  8790 non-null   int64
 7   rating        8790 non-null   object
```

```
 8   duration       8790 non-null   object
 9   listed_in      8790 non-null   object
dtypes: int64(1), object(9)
memory usage: 686.8+ KB
```

Visual representation of rating frequency of movies and TV Shows on Netflix.

In [9]:

```
data['rating'].value_counts()
```

Out[9]:

```
rating
TV-MA        3205
TV-14        2157
TV-PG         861
R             799
PG-13         490
TV-Y7         333
TV-Y          306
PG            287
TV-G          220
NR             79
G              41
TV-Y7-FV        6
NC-17           3
UR              3

Name: count, dtype: int64
```

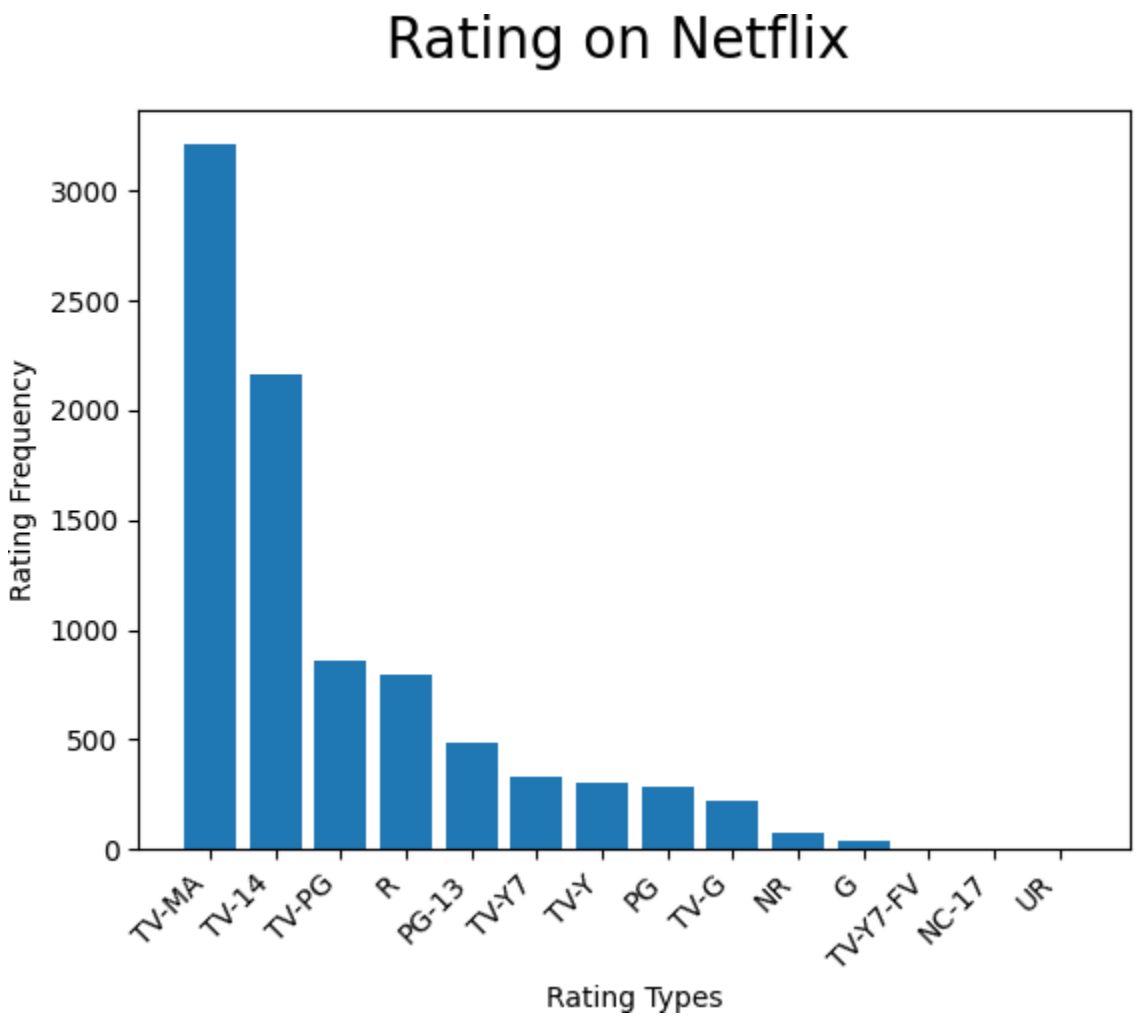
In [10]:

```
ratings=data['rating'].value_counts().reset_index().sort_values
(by='count', ascending=False)
```

```
plt.bar(ratings['rating'], ratings['count'])
plt.xticks(rotation=45, ha='right')
plt.xlabel("Rating Types")
plt.ylabel("Rating Frequency")

plt.suptitle('Rating on Netflix', fontsize=20)
```

Out[10]:

```
Text(0.5, 0.98, 'Rating on Netflix')
```

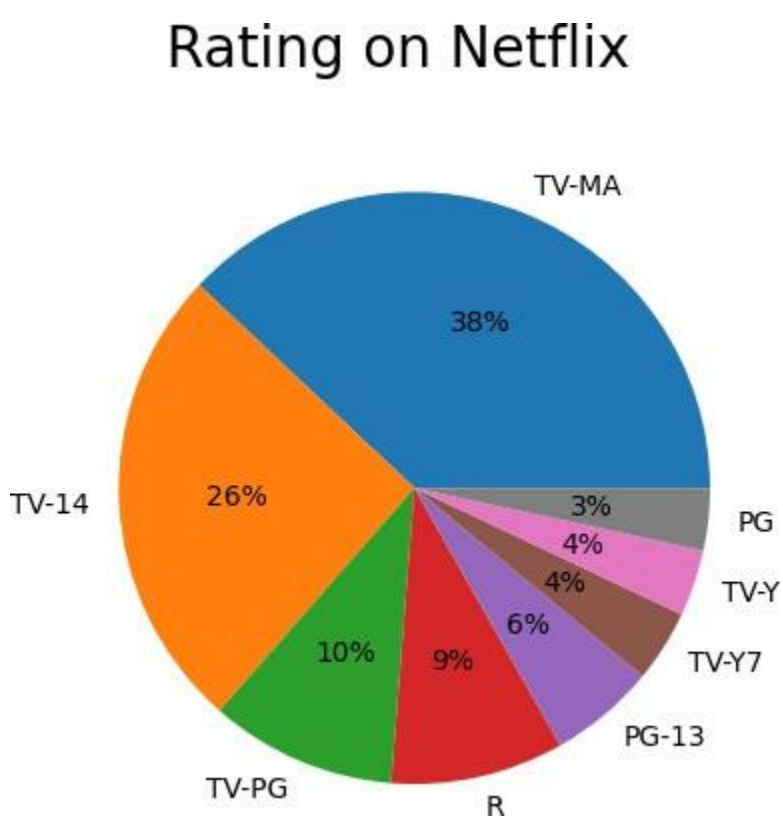


In [11]:

```
plt.pie(ratings['count'][:8], labels=ratings['rating'][:8],
autopct='%.0f%%')
plt.suptitle('Rating on Netflix', fontsize=20)
```

Out[11]:

```
Text(0.5, 0.98, 'Rating on Netflix')
```



Converting date_added column to datetime.

In [12]:

```
# lets convert column date_added to datetime.
data['date_added']=pd.to_datetime(data['date_added'])
```

In [13]:

```
data.describe()
```

Out[13]:

	date_added	release_year
count	8790	8790.000000
mean	2019-05-17 21:44:01.638225408	2014.183163
min	2008-01-01 00:00:00	1925.000000
25%	2018-04-06 00:00:00	2013.000000

50%	2019-07-03 00:00:00	2017.000000
75%	2020-08-19 18:00:00	2019.000000
max	2021-09-25 00:00:00	2021.000000
std	NaN	8.825466

In [14]:

```
data['country'].value_counts()
```

Out[14]:

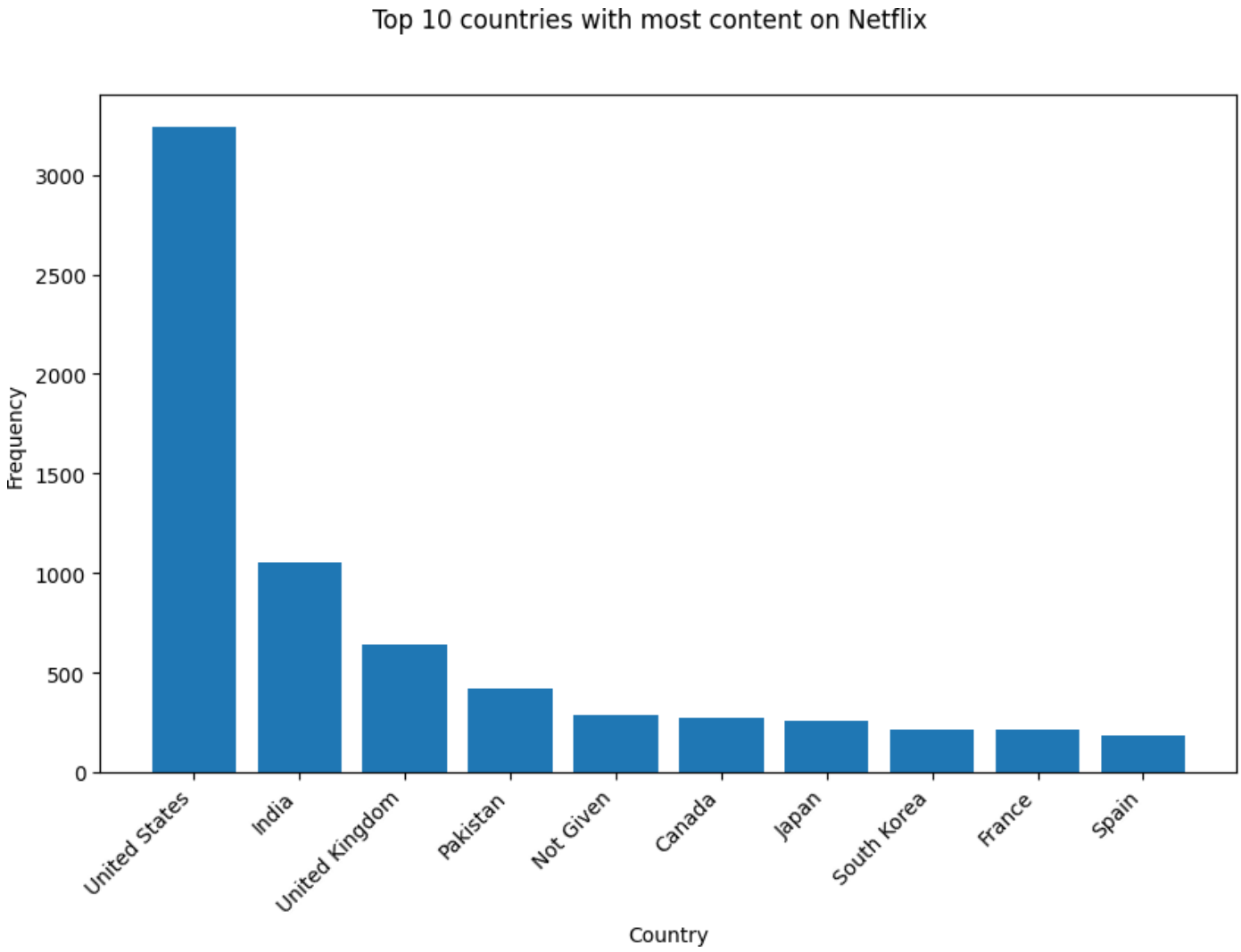
```
country
United States      3240
India              1057
United Kingdom      638
Pakistan            421
Not Given           287
                   ...
Iran                  1
West Germany          1
Greece                1
Zimbabwe              1
```

```
Soviet Union          1

Name: count, Length: 86, dtype: int64
```

Top 10 countries with most content on Netflix

In [15]:

```
top_ten_countries=data['country'].value_counts().reset_index().
sort_values(by='count', ascending=False)[:10]

plt.figure(figsize=(10, 6))
plt.bar(top_ten_countries['country'],
top_ten_countries['count'])
plt.xticks(rotation=45, ha='right')
plt.xlabel("Country")
plt.ylabel("Frequency")
plt.suptitle("Top 10 countries with most content on Netflix")
plt.show()
```



In [16]:

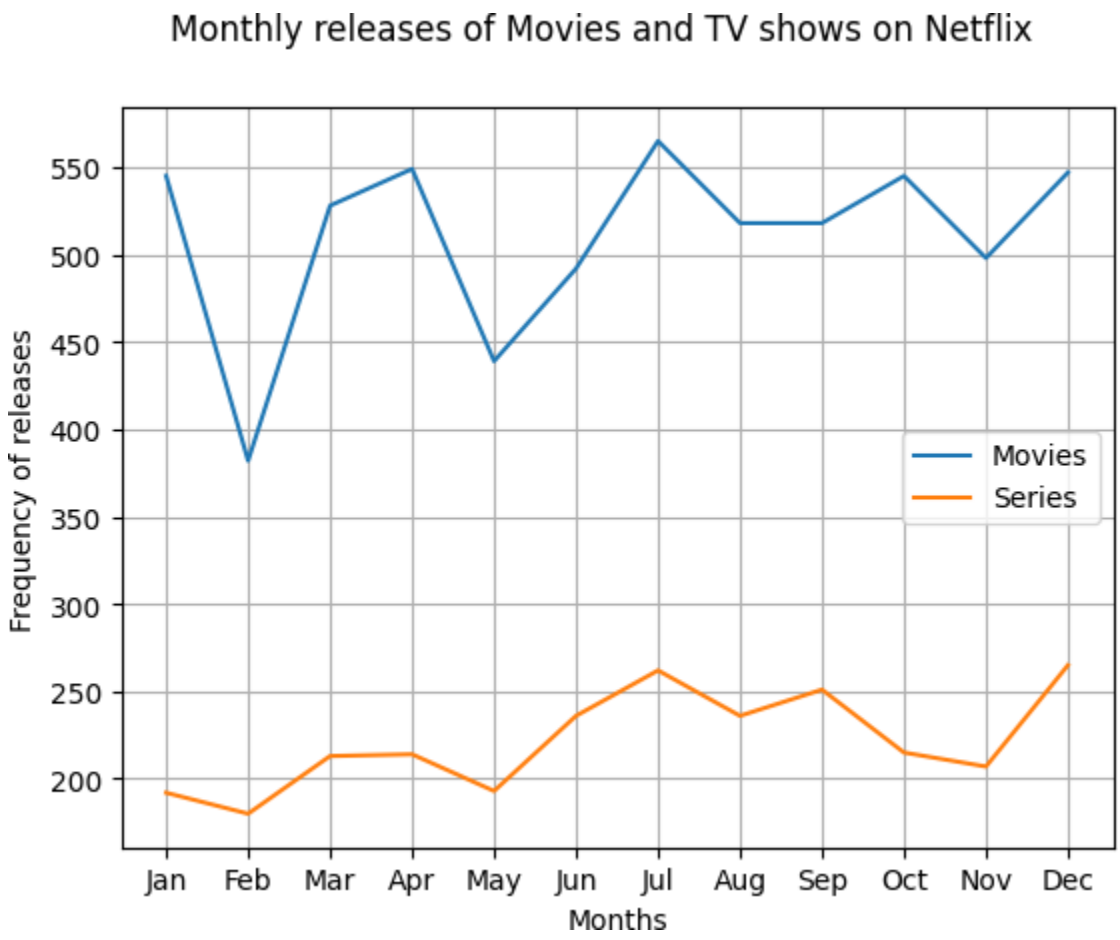
```
data['year']=data['date_added'].dt.year
data['month']=data['date_added'].dt.month
data['day']=data['date_added'].dt.day
```

Monthly releases of Movies and TV shows on Netflix

In [17]:

```
monthly_movie_release=data[data['type']=='Movie']['month'].valu
e_counts().sort_index()
monthly_series_release=data[data['type']=='TV
```

```
Show'][['month'].value_counts().sort_index()

plt.plot(monthly_movie_release.index,
monthly_movie_release.values, label='Movies')
plt.plot(monthly_series_release.index,
monthly_series_release.values, label='Series')
plt.xlabel("Months")
plt.ylabel("Frequency of releases")
plt.xticks(range(1, 13), ['Jan', 'Feb', 'Mar', 'Apr', 'May',
'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'])
plt.legend()
plt.grid(True)
plt.suptitle("Monthly releases of Movies and TV shows on
Netflix")
plt.show()
```



Yearly releases of Movies and TV Shows on Netflix

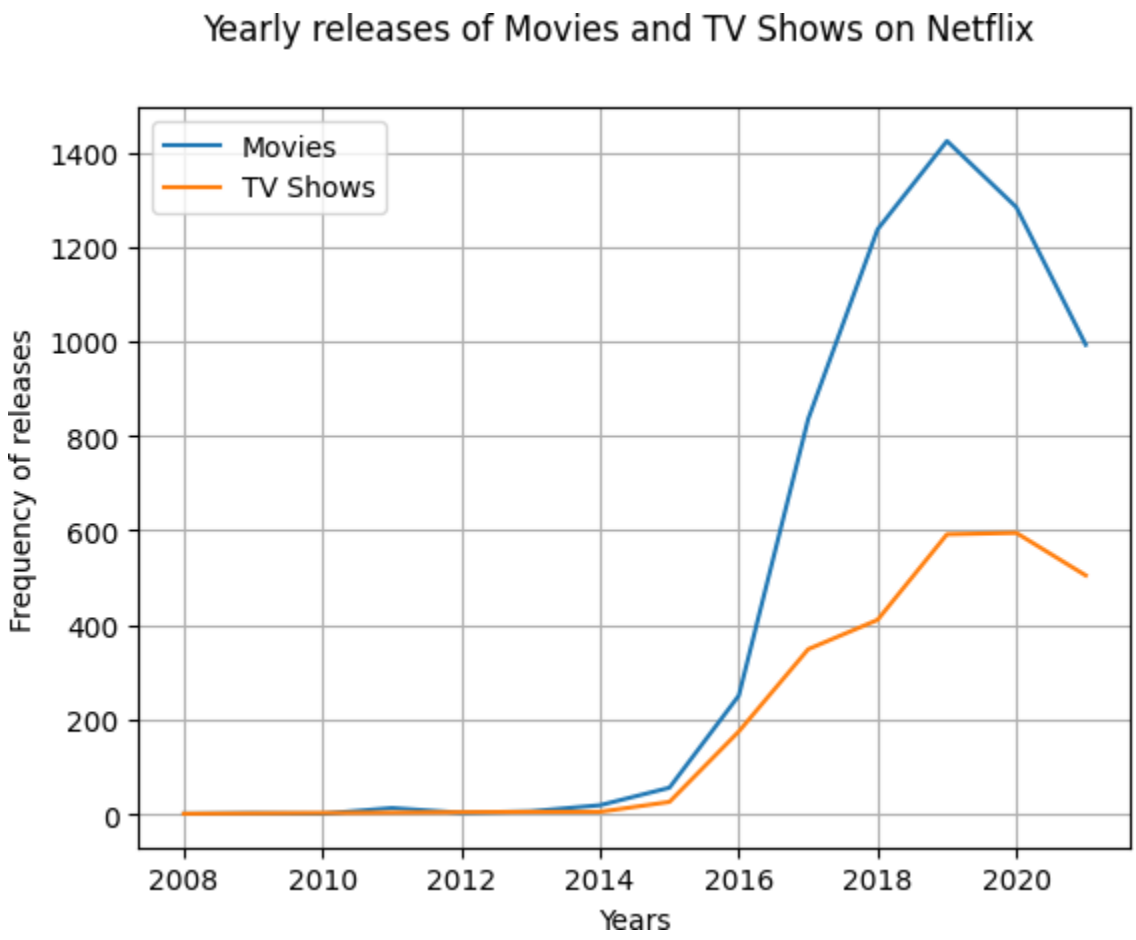
In [18]:

```
yearly_movie_releases=data[data['type']=='Movie']['year'].value
_counts().sort_index()
yearly_series_releases=data[data['type']=='TV
Show']['year'].value_counts().sort_index()

plt.plot(yearly_movie_releases.index,
yearly_movie_releases.values, label='Movies')
plt.plot(yearly_series_releases.index,
yearly_series_releases.values, label='TV Shows')
plt.xlabel("Years")
plt.ylabel("Frequency of releases")
plt.grid(True)
plt.suptitle("Yearly releases of Movies and TV Shows on
```

```
Netflix")
plt.legend()
```

Out[18]:

```
<matplotlib.legend.Legend at 0x7a14cb8327a0>
```

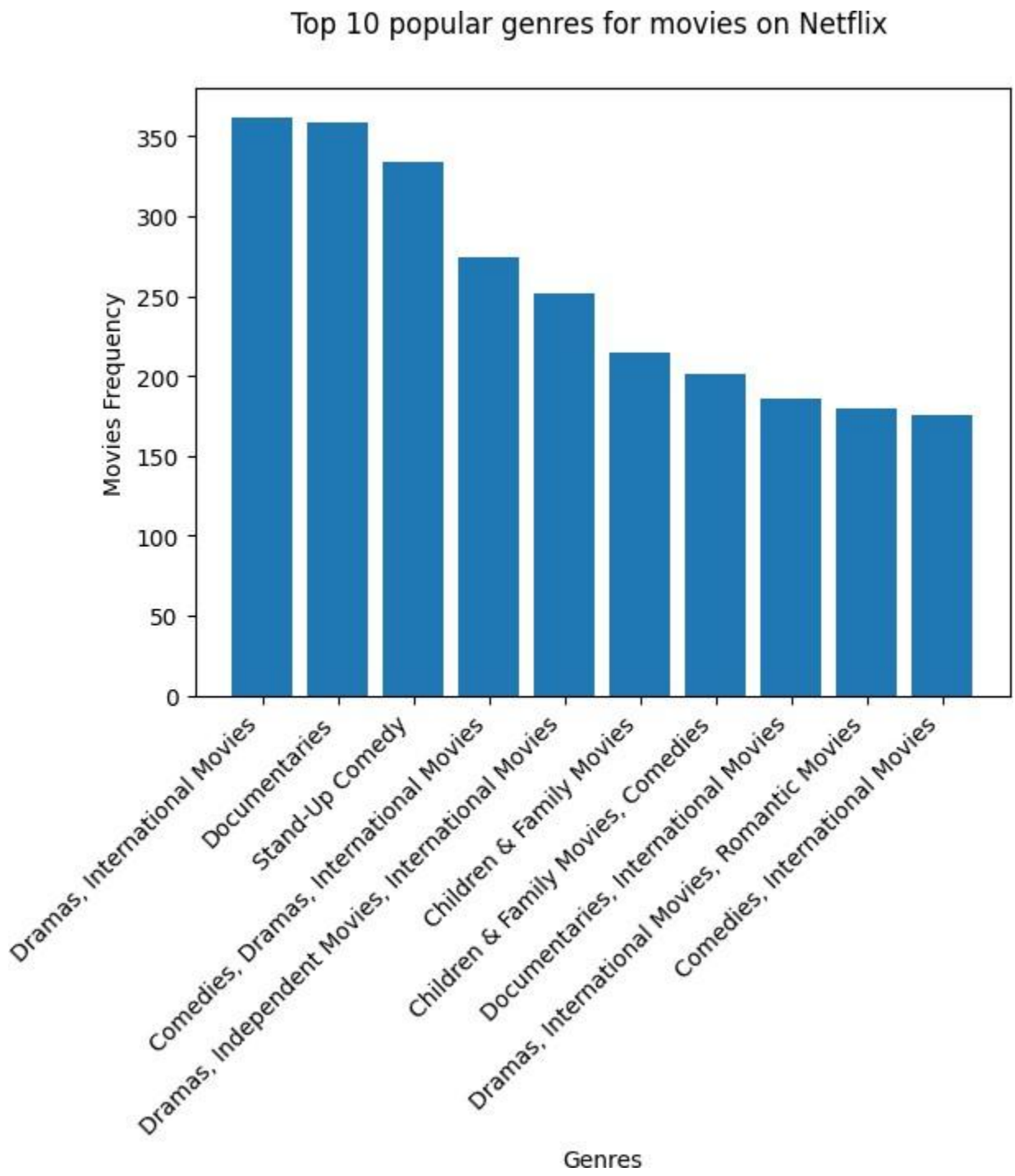


Top 10 popular movie genres

In [19]:

```
popular_movie_genre=data[data['type']=='Movie'].groupby("listed
_in").size().sort_values(ascending=False)[:10]
popular_series_genre=data[data['type']=='TV
```

```
Show'].groupby("listed_in").size().sort_values(ascending=False)
[:10]


plt.bar(popular_movie_genre.index, popular_movie_genre.values)
plt.xticks(rotation=45, ha='right')
plt.xlabel("Genres")
plt.ylabel("Movies Frequency")
plt.suptitle("Top 10 popular genres for movies on Netflix")
plt.show()
```

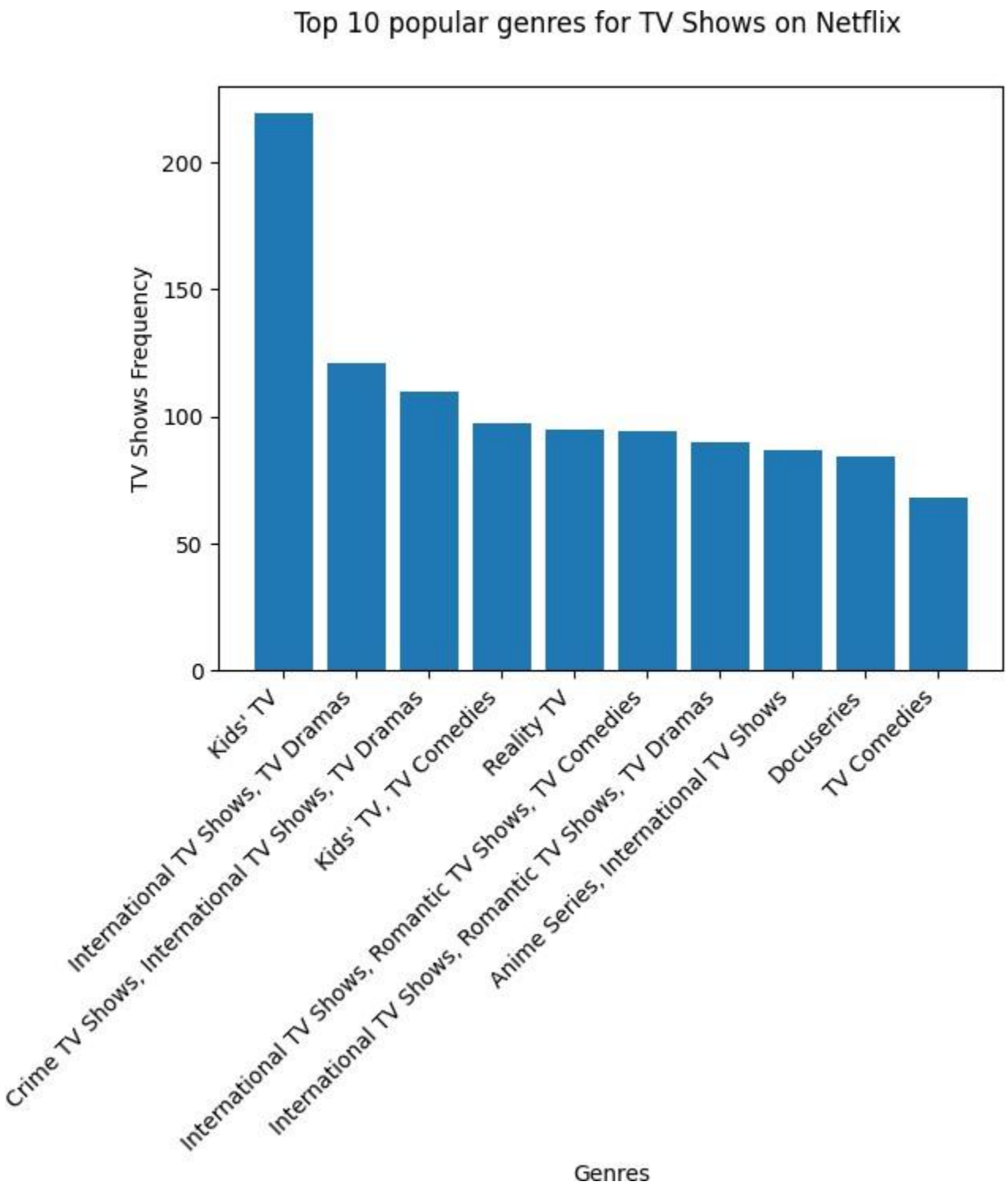


Top 10 TV Shows genres

In [20]:

```
plt.bar(popular_series_genre.index,
popular_series_genre.values)
plt.xticks(rotation=45, ha='right')
```

```
plt.xlabel("Genres")
plt.ylabel("TV Shows Frequency")
plt.suptitle("Top 10 popular genres for TV Shows on Netflix")
plt.show()
```



Top 15 directors across Netflix with hoigh frequency of movies and shows.

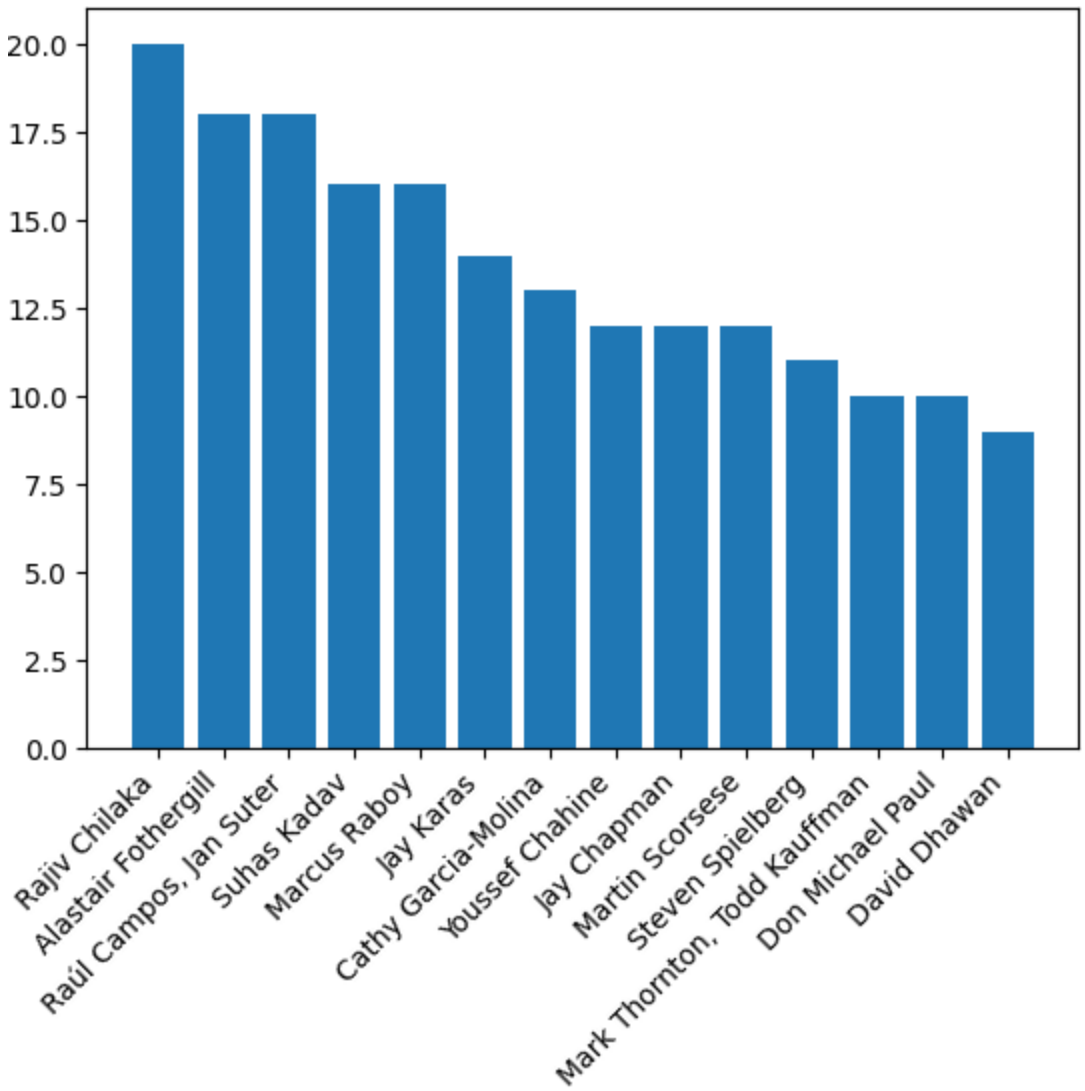
In [21]:

```
directors=data['director'].value_counts().reset_index().sort_va
```

```
lues(by='count', ascending=False)[1:15]

plt.bar(directors['director'], directors['count'])
plt.xticks(rotation=45, ha='right')
```

Out[21]:

```
([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13],
 [Text(0, 0, 'Rajiv Chilaka'),
  Text(1, 0, 'Alastair Fothergill'),
  Text(2, 0, 'Raúl Campos, Jan Suter'),
  Text(3, 0, 'Suhas Kadav'),
  Text(4, 0, 'Marcus Raboy'),
  Text(5, 0, 'Jay Karas'),
  Text(6, 0, 'Cathy Garcia-Molina'),
  Text(7, 0, 'Youssef Chahine'),
  Text(8, 0, 'Jay Chapman'),
  Text(9, 0, 'Martin Scorsese'),
  Text(10, 0, 'Steven Spielberg'),
  Text(11, 0, 'Mark Thornton, Todd Kauffman'),
  Text(12, 0, 'Don Michael Paul'),

  Text(13, 0, 'David Dhawan')])
```



In [ ]:

[Reference link]

[Reference link]